

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 11: Canny Edge Detection – Stage Analysis

PART 1: CORRECT ORDER OF STAGES

1. Gaussian Smoothing
2. Gradient Calculation
3. Non-Maximum Suppression
4. Double Thresholding

PART 2: QUESTIONS & SOLUTIONS

1. 1. What is the purpose of Non-Maximum Suppression?

The purpose of Non-Maximum Suppression (NMS) is to thin out thick edge responses by suppressing all gradient values that are not local maxima along the gradient direction, preserving only single-pixel-wide sharp edge lines.

2. 2. Why does Canny use double thresholding?

Canny uses double thresholding (high threshold T_{high} and low threshold T_{low}) to categorize candidate edge pixels into strong, weak, and non-edges. This preserves genuine weak edge pixels connected to strong edges (edge tracking by hysteresis) while eliminating isolated noise pixels that exceed a single low threshold.